

Statistical Questions and Data

Lesson 8-1

Name: _____

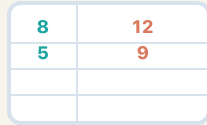
Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

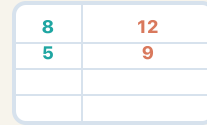


8	12
5	9

"How tall are the players on the team?" → answers vary: 60 in, 63 in, 58 in, 65 in

Statistical Question

A question where the answers will be different for different people.



8	12
5	9

Player heights: 60 in, 63 in, 58 in, 65 in — four different measurements

Data

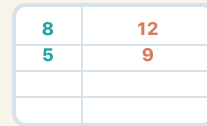
Facts and numbers you collect, like answers to a survey.



Scores 10, 12, 11 are close together (low variability); scores 2, 15, 30 are far apart (high variability)

Variability

How spread out the numbers are.

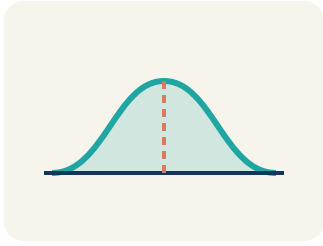


8	12
5	9

Asking 50 students 'What is your favorite sport?' and recording the answers

Survey

Collecting facts by asking people questions.



If most scores cluster in the middle with fewer at the edges, the distribution is symmetric

Data distribution

How the data looks: where it sits and how spread out it is.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can tell the difference between a statistical question and a non-statistical question.

1 Notice

You've just been hired as the data analyst for a youth basketball league.

2 Key idea

I can tell the difference between a statistical question and a non-statistical question.

3 Words I'll use

Statistical Question

Data

Variability

Survey

Data distribution



Think About It

- Which questions could have many different answers?
- Which questions have only one exact answer?
- What makes a question good for collecting data?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Which of the following is a statistical question?

- A. How many points did each player score this season?
- B. How many points are scored for a free throw?
- C. What sport does this team play?
- D. How many halves are in a soccer game?

Step 1

"How many points did each player score?" is statistical because different players will have different scores — there is variability in the data.



Answer: A. How many points did each player score this season?

 We do — together

Solve this one with your class using the same steps.

Problem: Which question is NOT a statistical question?

- A. How many innings are in a baseball game?
- B. How many home runs did each player hit this year?
- C. How many miles does each runner train per week?
- D. How tall is each player on the basketball team?

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Why is 'How many hours of sleep did each student get last night?' a statistical question?

- A. Because different students got different amounts of sleep
- B. Because it asks about sleep
- C. Because hours are numbers
- D. Because the answer is always 8 hours

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The coach asks, "How tall is our point guard?" and "How tall is each player on the roster?" Which one is a statistical question, and why?

Level 1 support

Start here: A statistical question expects answers that vary across many cases.

 **Need a hint?**

Sentence starters (optional)

The question "___" is statistical because ____.

La pregunta "___" es estadística porque ____.

The other question is not statistical because it has only ____ answer.

La otra pregunta no es estadística porque tiene solo ____ respuesta.

Word bank: statistical question variability data survey answer

Listen for: Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

Level 2

Take a non-statistical question and reword it so it becomes statistical. Explain exactly what you changed to create variability.

- I changed it to "___" so the answers will ____.
- Now it is statistical because the data will ____ from player to player.

EXPLORE

While sorting the questions, what clue tells you instantly that a question will need DATA from many players to answer?

Level 1 support

Start here: Questions that ask about "each" player usually anticipate variability.



Need a hint?

Sentence starters (optional)

I know it needs data because it asks about ____.

Sé que necesita datos porque pregunta sobre ____.

The answers will be different because ____.

Las respuestas serán diferentes porque ____.

Word bank: **statistical question** **variability** **data** **survey** **vary**

Listen for: A strong answer notes words like "each" or "per player" signal variability, so you must collect data from many cases rather than look up one fact.

Level 2

Is it possible for a question with a numerical answer to still be NOT statistical? Justify with an example from the basketball league.

- Yes, a question like "____" has numbers but is not statistical because ____.
- It is not statistical even with numbers since the answer ____.

CONNECT

The newspaper editor prefers "How many goals did each player score?" over "How many goals did the team score total?" Which gives more interesting data for a story?

Level 1 support

Start here: A statistical question reveals variability that a single total hides.



Need a hint?

Sentence starters (optional)

The editor's question is better because it shows ____ among players.

La pregunta del editor es mejor porque muestra ____ entre los jugadores.

The total only gives ____ number, so it hides ____.

El total solo da ____ número, así que oculta ____.

Word bank: (statistical question) (variability) (data) (survey) (total)

Listen for: Students explain the per-player question is statistical and shows variability (who scored most/least), while the total is one fixed number that hides individual differences.

Level 2

Critique the reporter's plan: when might a single total actually be the RIGHT thing to report instead of per-player data? Defend your view.

- A total would be better when the reader wants to know ____.
- But per-player data is better here because the story is about ____.

REFLECT

Think of one statistical question you could ask about students in this room. What variability do you expect in the answers?

Level 1 support

Start here: A good statistical question is answered by collecting varied data, not one fact.

 Need a hint?

Sentence starters (optional)

My statistical question is " ____ ".

Mi pregunta estadística es " ____ ".

I expect variability because ____.

Espero variabilidad porque ____.

Word bank: statistical question variability data survey collect

Listen for: Students pose a question whose answers vary across classmates and name the variability they expect (e.g., different bedtimes, different shoe sizes).

Level 2

Generalize: what is the one test you can always run on any question to decide if it is statistical? State your rule in one sentence.

- My rule is: a question is statistical if ____.
- I can test any question by asking whether the answers will ____.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words statistical question, data, variability, and survey.

1. Kernel Sentence subject + verb

Model: Statistical Question is a question where the answers will be different for different people.
Pregunta estadística es una pregunta cuyas respuestas serán distintas para distintas personas.

Write a kernel sentence about statistical question. Use a subject and a verb.

Escribe una oración base sobre pregunta estadística. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Statistical Question matters in math
Pregunta estadística importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Statistical Question matters in math because ____.

Pregunta estadística importa en matemáticas porque ____.

but
pero

Statistical Question matters in math, but ____.

Pregunta estadística importa en matemáticas, pero ____.

so
entonces

Statistical Question matters in math, so ____.

Pregunta estadística importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about statistical question.
Di un hecho verdadero sobre statistical question.

Statistical question ____.

Question *Pregunta*

Ask a question about statistical question.
Haz una pregunta sobre statistical question.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about statistical question.
Muestra entusiasmo sobre statistical question.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with statistical question.
Dile a un compañero qué hacer con statistical question.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The question " ____ " is statistical because ____.

La pregunta " ____ " es estadística porque ____.

The other question is not statistical because it has only ____ answer.

La otra pregunta no es estadística porque tiene solo ____ respuesta.

I know it needs data because it asks about ____.

Sé que necesita datos porque pregunta sobre ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Why do statistical questions expect different answers?

- A. Because the data values vary from one case to another
- B. Because they are always about numbers
- C. Because they cannot be answered
- D. Because they only have one answer

Show your work:

2. Which rewrite makes 'How old am I?' a statistical question?

- A. How old are the students in this school?
- B. How old is my best friend?
- C. When is my birthday?
- D. Am I older than 12?

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write one statistical question and one non-statistical question about your school's cafeteria. Explain why each question is or is not statistical, and describe what the data would look like for the statistical question.

Sentence starter: Statistical question: _____. This is statistical because _____. The data might look like: _____.

Non-statistical question: _____. This is not statistical because _____.

Show your work:

Reflect — Exit Ticket

Which question is a statistical question? A) How many students are in the school? B) How many hours does each student in 6th grade exercise per week? C) What time does school start? D) How many days are in a week?

A. B) How many hours does each student exercise per week?

B. A) How many students are in the school?

C. C) What time does school start?

D. D) How many days are in a week?

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. How many innings are in a baseball game? — *"How many innings are in a baseball game?" has one fixed answer (9) — no variability. The other questions all produce different answers for different people.*
2. **You Try (notes):** A. Because different students got different amounts of sleep — *A statistical question anticipates variability — different students slept different amounts, so the data will vary.*
3. **Try It 1:** A. Because the data values vary from one case to another — *Statistical questions anticipate variability in the data.*
4. **Try It 2:** A. How old are the students in this school? — *Asking about many students produces data that varies.*
5. **Exit Ticket:** A. B) How many hours does each student exercise per week? — *"How many hours does each student exercise per week?" is statistical because different students exercise different amounts — the data has variability.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Statistical Question is a question where the answers will be different for different people.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).